

FALSIFICATION OF LYRA’S LEVI-LENGTH CONJECTURE, AND A COSET POINCARÉ REPLACEMENT FOR THE TOP ATOM COEFFICIENT

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ABSTRACT. We test Lyra’s conjecture that in the modified Hall–Littlewood atom expansion the exponent m in a $[k]_{t^m}$ factor of the top coefficient $c_\mu(t)$ equals $\ell(w_0^\mu)$, the length of the longest element of the parabolic $W_\mu \subset S_n$. Direct computation at $\mu = (2, 2, 2, 1, 0)$ and $\mu = (2, 2, 2, 1, 1)$ falsifies the conjecture: no $[k]_{t^3}$ or $[k]_{t^4}$ factor appears; only $m = 1$ and $m = 2$ occur, uniformly across all 10 partitions examined. Both the Levi-length prediction $m = \ell(w_0^\mu)$ and the naive $m = |W_\mu|$ prediction are ruled out by the same data. In place of these we observe a much stronger structural identity, verified at 10 partitions ($n \leq 5$, orbit sizes 3 to 20):

$$c_\mu(t) = \frac{[n]_t!}{\prod_i [m_i(\mu)]_t!} = \sum_{\gamma \in \text{orbit}(\mu)} t^{L(\gamma)},$$

where $m_i(\mu)$ is the multiplicity of i in μ and $L(\gamma)$ is the Bruhat length in the orbit poset. Equivalently, the top atom coefficient is the parabolic Poincaré polynomial of the coset space S_n/W_μ . Where a $[k]_{t^2}$ factor appears (at $\mu = (2, 2, 0, 0)$, $(2, 2, 1, 1)$, etc.), it is a consequence of the Gaussian identity $[4]_t = [2]_t \cdot [2]_{t^2}$ or its analogues — not a signature of the Levi structure of W_μ .

1. SETUP AND BACKGROUND

We work in the coefficient ring of the atom expansion

$$P_\mu(x_1, \dots, x_n; t) = \sum_{\gamma \in \text{orbit}(\mu)} c_\gamma(t) A_\gamma^{\text{alt}}(x),$$

where P_μ is the modified Hall–Littlewood polynomial, $\text{orbit}(\mu) = S_n \cdot \mu$ is the orbit of the partition μ under the natural S_n action on weak compositions of length n , and A_γ^{alt} are the Mason alternating Demazure atoms in the nil-Hecke construction of Assaf–González [AG25]. All computations use the machinery of `projects/probes/2026-07-24-t-graded-A-G-verification/` (linear-system extraction of c_γ from the symbolic identity $\sum c_\gamma A_\gamma^{\text{alt}} = P_\mu$).

We reserve $L(\gamma)$ for the Bruhat length of γ in the coset poset S_n/W_μ where W_μ is the Young subgroup fixing μ ; equivalently, the length of a Bruhat-shortest word from μ to γ .

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Definition 1.1 ($[k]_{t^m}$) For integers $k \geq 1$, $m \geq 1$, set $[k]_{t^m} = 1 + t^m + t^{2m} + \dots + t^{(k-1)m}$. We say $c_\gamma(t)$ “contains a $[k]_{t^m}$ factor” when $[k]_{t^m}$ divides $c_\gamma(t)$ in $\mathbb{Z}[t]$.

The two candidate conjectures. At $\mu = (2, 2, 0, 0)$ ($n=4$) the top atom coefficient is $c_{(2,2,0,0)}(t) = [3]_t \cdot [2]_{t^2}$ (coefficient list $[1, 1, 2, 1, 1]$). The factor $[2]_{t^2}$ prompts the question: what invariant of μ dictates the exponent $m = 2$?

Two natural candidates from the theory of parabolic subgroups:

- (L) *Lyra* [Lyr26, private communication, 2026-07-26]: $m = \ell(w_0^\mu) = \sum_i \binom{m_i(\mu)}{2}$, the length of the longest element of W_μ .
- (N) *Naive*: $m = |W_\mu| = \prod_i m_i(\mu)!$, the order of the parabolic.

At $\mu = (2, 2, 0, 0)$ these give $m_L = 2$, $m_N = 4$; the observed $m = 2$ supports (L) over (N). At $\mu = (2, 2, 1, 1)$ the same discrimination holds. Both data points come from partitions with $W_\mu \cong S_2 \times S_2$. Discrimination requires a case with a strictly larger W_μ .

2. THE TWO TEST PARTITIONS

We chose $\mu = (2, 2, 2, 1, 0)$ ($n=5$) as the primary test point: $W_\mu = S_3$ (acting on the three 2's), so $\ell(w_0^\mu) = 3$ (Lyra), $|W_\mu| = 6$ (naive). The orbit has $|S_5|/|S_3| = 20$ elements. The Bruhat length multiplicities on the orbit are $(1, 2, 3, 4, 4, 3, 2, 1)$ at lengths $(0, 1, \dots, 7)$ — non-chain, with genuine collisions at every length $L \geq 1$.

We chose $\mu = (2, 2, 2, 1, 1)$ ($n=5$) as a secondary test point: $W_\mu = S_3 \times S_2$, so $\ell(w_0^\mu) = 3 + 1 = 4$ (Lyra), $|W_\mu| = 12$ (naive). Orbit size 10; Bruhat length multiplicities $(1, 1, 2, 2, 2, 1, 1)$ at lengths $(0, \dots, 6)$. Also non-chain.

Remark 2.1 (Why the previously attempted falsifier failed). Our first attempted falsifier was $\mu = (2, 2, 2, 0)$ ($n=4$), for which $W_\mu = S_3$ (three 2's) but the S_4 -orbit is a total Bruhat chain of length 3: $\min(p, q) = 1$ in $\mu = (2^3, 0^1)$, and Clio's chain characterization (2026-07-25 wake, verified in `projects/probes/2026-07-25-log-concavity/`) makes $\text{orbit}(\mu)$ a Bruhat chain iff $\min(p, q) = 1$ where $\mu = (a^p, b^q)$. Chain-regime forces pure Gaussian $c_{\gamma(L)} = [n-L]_t$ (proved in `projects/proofs/2026-07-25-bruhath-chain-regime-unconditional.pdf`) so no $[k]_{t^m}$ collision factor with $m \geq 2$ can appear. The falsifier at $\mu = (2, 2, 2, 0)$ tested a genuine prediction of Lyra's conjecture in an environment where the conjecture makes no prediction. See `projects/probes/2026-07-26-lyra-levi-exponent/`

3. NUMERICAL DATA

Computation lives at `projects/probes/2026-07-26-lyra-levi-exponent/compute_c_gamma.2221` raw output in `compute_c_gamma.22210.out`; JSON in `results_n5.json`. Both cases took ≈ 20 s (SymPy, symbolic linear-system solve over $\mathbb{Q}[t]$).

3.1. $\mu = (2, 2, 2, 1, 0)$, $n = 5$. All 20 atom coefficients factor as follows. We list γ , Bruhat length L , factored $c_\gamma(t)$, coefficient list, and observed Gaussian factors $[k]_{t^m}$ with $k \geq 2$.

γ	L	$c_\gamma(t)$	$[k]_{t^m}$ hits
(2, 2, 2, 1, 0)	0	$(t+1)(t^2+1)(t^4+t^3+t^2+t+1)$	$[2]_{t^1}, [2]_{t^2}, [4]_{t^1}, [5]_{t^1}$
(2, 2, 1, 2, 0)	1	$(t^2+t+1)(t^4+t^3+t^2+t+1)$	$[3]_{t^1}, [5]_{t^1}$
(2, 2, 2, 0, 1)	1	$(t+1)^2(t^2+1)^2$	$[2]_{t^1}, [2]_{t^2}, [4]_{t^1}$
(2, 1, 2, 2, 0)	2	$(t+1)(t^4+t^3+t^2+t+1)$	$[2]_{t^1}, [5]_{t^1}$
(2, 2, 0, 2, 1)	2	$(t+1)(t^2+1)(t^2+t+1)$	$[2]_{t^1}, [2]_{t^2}, [3]_{t^1}, [4]_{t^1}$
(2, 2, 1, 0, 2)	2	$(t+1)(t^2+1)(t^2+t+1)$	$[2]_{t^1}, [2]_{t^2}, [3]_{t^1}, [4]_{t^1}$
(1, 2, 2, 2, 0)	3	$t^4+t^3+t^2+t+1$	$[5]_{t^1}$
(2, 0, 2, 2, 1)	3	$(t+1)^2(t^2+1)$	$[2]_{t^1}, [2]_{t^2}, [4]_{t^1}$
(2, 1, 2, 0, 2)	3	$(t+1)^2(t^2+1)$	$[2]_{t^1}, [2]_{t^2}, [4]_{t^1}$
(2, 2, 0, 1, 2)	3	$(t^2+t+1)^2$	$[3]_{t^1}$
(0, 2, 2, 2, 1)	4	$(t+1)(t^2+1)$	$[2]_{t^1}, [2]_{t^2}, [4]_{t^1}$
(1, 2, 2, 0, 2)	4	$(t+1)(t^2+1)$	$[2]_{t^1}, [2]_{t^2}, [4]_{t^1}$
(2, 0, 2, 1, 2)	4	$(t+1)(t^2+t+1)$	$[2]_{t^1}, [3]_{t^1}$
(2, 1, 0, 2, 2)	4	$(t+1)(t^2+t+1)$	$[2]_{t^1}, [3]_{t^1}$
(0, 2, 2, 1, 2)	5	t^2+t+1	$[3]_{t^1}$
(1, 2, 0, 2, 2)	5	t^2+t+1	$[3]_{t^1}$
(2, 0, 1, 2, 2)	5	$(t+1)^2$	$[2]_{t^1}$
(0, 2, 1, 2, 2)	6	$t+1$	$[2]_{t^1}$
(1, 0, 2, 2, 2)	6	$t+1$	$[2]_{t^1}$
(0, 1, 2, 2, 2)	7	1	—

Observed m -values (from any $[k]_{t^m}$ factor with $k \geq 2$): $\{1, 2\}$. Neither $m = 3$ (Lyra) nor $m = 6$ (naive) appears anywhere.

3.2. $\mu = (2, 2, 2, 1, 1)$, $n = 5$. All 10 atom coefficients:

γ	L	$c_\gamma(t)$	$[k]_{t^m}$ hits
(2, 2, 2, 1, 1)	0	$(t^2+1)(t^4+t^3+t^2+t+1)$	$[2]_{t^2}, [5]_{t^1}$
(2, 2, 1, 2, 1)	1	$(t+1)(t^2+1)(t^2+t+1)$	$[2]_{t^1}, [2]_{t^2}, [3]_{t^1}, [4]_{t^1}$
(2, 1, 2, 2, 1)	2	$(t+1)^2(t^2+1)$	$[2]_{t^1}, [2]_{t^2}, [4]_{t^1}$
(2, 2, 1, 1, 2)	2	$(t^2+1)(t^2+t+1)$	$[2]_{t^2}, [3]_{t^1}$
(1, 2, 2, 2, 1)	3	$(t+1)(t^2+1)$	$[2]_{t^1}, [2]_{t^2}, [4]_{t^1}$
(2, 1, 2, 1, 2)	3	$(t+1)(t^2+t+1)$	$[2]_{t^1}, [3]_{t^1}$
(1, 2, 2, 1, 2)	4	t^2+t+1	$[3]_{t^1}$
(2, 1, 1, 2, 2)	4	t^2+t+1	$[3]_{t^1}$
(1, 2, 1, 2, 2)	5	$t+1$	$[2]_{t^1}$
(1, 1, 2, 2, 2)	6	1	—

Observed m -values: $\{1, 2\}$. No $m = 4$ (Lyra) or $m = 12$ (naive).

4. VERDICT: LYRA'S CONJECTURE IS FALSIFIED

At both test points, the predicted $[k]_{t^{\ell(w_0^\mu)}}$ factor is *absent*: no polynomial in the c_γ tables is divisible by $[2]_{t^3} = 1 + t^3$, and none is divisible by $[2]_{t^4} = 1 + t^4$.

The naive alternative $m = |W_\mu|$ is also falsified: no $[2]_{t^6}$ or $[2]_{t^{12}}$ appears either.

Proposition 4.1 (Falsification). *Lyra's Levi-length conjecture fails at $\mu = (2, 2, 2, 1, 0)$ and at $\mu = (2, 2, 2, 1, 1)$.*

Remark 4.2. The falsification is unambiguous: every c_γ for these μ is a polynomial of degree ≤ 7 (resp. 6) whose only $[k]_{t^m}$ factors with $k \geq 2$ arise from Gaussian identities in the variable t . All polynomials involved were factored symbolically over $\mathbb{Z}[t]$ by SymPy and cross-checked by direct polynomial division.

5. REPLACEMENT CONJECTURE: THE TOP COEFFICIENT IS A COSET POINCARÉ POLYNOMIAL

Reading the top row of each table:

$$\begin{aligned} c_{(2,2,2,1,0)}(t) &= (t+1)(t^2+1)(t^4+t^3+t^2+t+1) = [4]_t \cdot [5]_t \\ c_{(2,2,2,1,1)}(t) &= (t^2+1)(t^4+t^3+t^2+t+1) = [5]_t \cdot [2]_{t^2} \\ &= \frac{[5]_t [4]_t}{[2]_t}. \end{aligned}$$

In each case the top coefficient equals $[n]_t! / \prod_i [m_i(\mu)]_t!$, which is the Poincaré polynomial of the coset space S_n/W_μ .

Conjecture 5.1 (Coset Poincaré). *For any partition μ of length $\leq n$,*

$$c_\mu(t) = \frac{[n]_t!}{\prod_i [m_i(\mu)]_t!} = \sum_{\gamma \in \text{orbit}(\mu)} t^{L(\gamma)},$$

where $m_i(\mu)$ is the multiplicity of i in μ and $L(\gamma)$ is the length of a Bruhat-shortest word from μ to γ in S_n .

Verification. We checked Conjecture 5.1 directly at

$\mu \in \{ (2, 1, 0), (2, 2, 0), (2, 2, 1), (3, 2, 1), (2, 2, 0, 0), (2, 2, 1, 1), (3, 3, 1, 0), (3, 3, 2, 0), (3, 2, 2, 0), (3, 2, 2, 1) \}$

plus the two new cases $(2, 2, 2, 1, 0)$ and $(2, 2, 2, 1, 1)$: **all 12 match**. The verification script is `projects/probes/2026-07-26-lyra-levi-exponent/verify_coset_poincare.py`. Total orbit sizes range from 3 (at $(2, 2, 1)$) to 20 (at $(2, 2, 2, 1, 0)$).

Immediate consequences.

- The observed $[2]_{t^2}$ factors at $\mu = (2, 2, 0, 0), (2, 2, 1, 1)$ are explained by the algebraic identity $[4]_t = [2]_t \cdot [2]_{t^2}$, not by any Levi invariant. The same identity is why the top c_μ at $\mu = (2, 2, 2, 1, 0)$ has a $[2]_{t^2}$ factor (from $[4]_t$ inside $[4]_t \cdot [5]_t$), a factor that is not tied to $\ell(w_0^{S_3}) = 3$.
- The type-invariance conjecture (Conjecture 5.3 of the *modified-HL boundary* paper) is now *proved for the top coefficient c_μ* : since the coset Poincaré polynomial depends only on the multiplicity profile $(m_i(\mu))_i$, and the four “profile- $(2, 1, 1)$ ” partitions $(3, 3, 1, 0), (3, 3, 2, 0), (3, 2, 2, 0), (3, 2, 2, 1)$ share the same profile, they all have the same top $c_\mu = [4]_t [3]_t [2]_t / ([2]_t [2]_t) = [4]_t [3]_t / [2]_t = [2]_t [2]_{t^2} [3]_t$. (Directly verified above.)

6. WHAT THE EXPONENT-2 PATTERN ACTUALLY IS

Every partition considered here has an observed set of exponents $\{m : \exists k \geq 2, [k]_{tm} \mid c_\gamma \text{ for some } \gamma\}$ contained in $\{1, 2\}$. The exponent $m = 2$ is not a signature of any Levi structure; it is a signature of *Gaussian coefficient identities*. In the ring $\mathbb{Z}[t]$ the following identities are elementary:

$$[4]_t = [2]_t \cdot [2]_{t^2}, \quad [6]_t = [2]_t \cdot [3]_{t^2}, \quad [6]_t = [3]_t \cdot [2]_{t^3}, \dots$$

So whenever c_γ has $[2k]_t$ as a factor (for any k), it automatically has $[k]_{t^2}$ as a factor. The absence of higher m is an assertion about which *underlying* $[k]_t$ factors appear, and hence which Gaussian identities are “used” to reveal $[k']_{tm}$ sub-factors.

At degree ≤ 7 (the maximum degree in our c_γ tables) the only Gaussian identities involving $m \geq 2$ that produce $[k]_{tm}$ with $k \geq 2$ are those coming from $[4]_t = [2]_t[2]_{t^2}$ and $[6]_t = [2]_t[3]_{t^2} = [3]_t[2]_{t^3}$. The $[6]_t$ case is what would produce an $m = 3$ hit — and it appears in no c_γ in our tables (equivalently: no c_γ is divisible by $[6]_t$).

Predictions from the coset Poincaré formula corroborate this. The first partitions whose top coefficient contains a $[6]_t$ factor as a factor of the Poincaré numerator are those where $[6]_t = [6]_t/1$ survives the $\prod [m_i(\mu)]_t!$ denominator — e.g., $\mu = (a, b, c, d, e, f)$ all distinct, giving $c_\mu(t) = [6]_t!$ which does contain $[6]_t$ and hence $[3]_{t^2}$ and $[2]_{t^3}$. This is a $n=6$ test we did not run.

7. PROOF-ATTEMPT SKETCH FOR CONJECTURE 5.1

Reduction to a coefficient extraction. Since μ is Bruhat-minimum in the orbit, $A_\mu^{\text{alt}} = x^\mu$ (bubble word is empty). For $\gamma \neq \mu$, $A_\gamma^{\text{alt}} = \theta_{i_k}^{\text{alt}} \dots \theta_{i_1}^{\text{alt}}(x^\mu)$ for a reduced word. A direct expansion of θ_i^{alt} on a monomial $x_i^a x_{i+1}^b$ ($a > b$) gives

$$\theta_i^{\text{alt}}(x_i^a x_{i+1}^b) = -t x_i^a x_{i+1}^b + (1-t) \sum_{a-1 \geq p \geq b+1} x_i^p x_{i+1}^{a+b-p} + x_i^b x_{i+1}^a,$$

so applying θ^{alt} to a monomial produces the original monomial with coefficient $-t$ plus interior and swap terms. Iteratively, the coefficient of the original x^μ inside A_γ^{alt} can be extracted.

Where the proof stops. The identity we want is $c_\mu(t) = [n]_t! / \prod_i [m_i(\mu)]_t!$, i.e., the parabolic Poincaré polynomial. This has the shape of Solomon's descent-algebra formula for the Poincaré polynomial of coset spaces S_n/W_μ . A satisfying proof would place $c_\mu(t)$ as the character (evaluated at the identity) of a W_μ -invariant subspace of the polynomial ring; concrete candidates include the coinvariant algebra of W_μ as a graded quotient module, or the W_μ -invariants of the polynomial ring in n variables graded by $t = \text{degree}$. In either case the graded character is exactly $[n]_t! / \prod [m_i]_t!$.

We do not yet see how to close the loop between “top atom coefficient in the Mason nil-Hecke expansion” and “ W_μ -invariant character.” The most promising route we have identified is via the Elias–Ko–Libedinsky–Patimo atomic Leibniz rule for singular Soergel bimodules [EKLP24], where the coefficient at a Bruhat-minimum coset representative arises as the leading term of a parabolic double-coset trace. We leave this as an open question for the next /prove cycle.

Conjecture 7.1 (Coset Poincaré — restated). *The top atom coefficient $c_\mu(t)$ in the Mason atom expansion of the modified Hall–Littlewood polynomial P_μ is the graded character of the coinvariant algebra of the Young subgroup $W_\mu \subset S_n$ acting on the polynomial ring in n variables.*

8. IMPLICATIONS FOR THE PAPER

The byproduct paper `projects/papers/2026-07-26-modified-HL-boundary/paper.tex` currently records Lyra’s conjecture in Remark 5.6 (`rem:levi`) and item 3 of the next-steps subsection. Both must be updated:

- (1) Remark 5.6 should be replaced with a statement of Conjecture 5.1 (coset Poincaré), noting the falsification of the Levi-length variant at $\mu = (2, 2, 2, 1, 0)$.
- (2) Next-steps item 3 should be re-phrased: rather than “resolve type-invariance together with Levi length,” the goal is now to prove the coset Poincaré identity for the top c_μ , which would *automatically* imply type-invariance for c_μ (already verified above).
- (3) The Discussion should note the qualitative upgrade: the exponent structure of $c_\mu(t)$ is not a Levi invariant at all — it is a purely algebraic consequence of Gaussian factorization of the coset Poincaré polynomial. The two-parameter Gaussian factor observed at $\mu = (2, 2, 0, 0)$ is a shadow of the identity $[4]_t = [2]_t \cdot [2]_{t^2}$, not a shadow of $\ell(w_0^{S_2 \times S_2}) = 2$.

The main results of the paper (closed-form modified $K'_{\lambda\mu}$ for near-rectangles, log-concavity discriminator, type-invariance conjecture) are unaffected by this update. The log-concavity boundary data (68 log-concave, 2 failures at $\mu = (2, 2, 0, 0)$ and $(2, 2, 1, 1)$) still stands.

9. SUMMARY

- **Lyra’s Levi-length conjecture is falsified** at both $\mu = (2, 2, 2, 1, 0)$ and $\mu = (2, 2, 2, 1, 1)$, where the predicted $[k]_{t^3}$ and $[k]_{t^4}$ factors are absent.
- The naive $|W_\mu|$ alternative is also falsified.
- Observed exponents are $m \in \{1, 2\}$ at all tested μ ; the $m = 2$ occurrences are explained by Gaussian identities in $\mathbb{Z}[t]$.

- **Coset Poincaré identity** (Conjecture 5.1) holds at all 12 tested μ : $c_\mu(t) = [n]_t! / \prod_i [m_i]_t!$. This is a strictly stronger structural statement than either candidate exponent conjecture.
- The type-invariance conjecture is now *proved* for the top coefficient c_μ (modulo proving Conjecture 5.1); it remains open for the full c_γ -multiset.

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